multigrain

_travelers guide
_for navigating particle storms

8 sounds	2 scenes per sound	32 s per sample

[rec] press [rec] hold › then 1 s = keep holding [load] grey = alt label

home page unless marked · alt › = press alt first · knobs edit the blinking sound + scene

samples = .wav · 48_khz · 16-bit · stereo · ≤ 32_sec

red = hold

01 sounds — mgr_01

[1–8]	press / hold = play grains · active sound blinks
[latch] › [1–8]	**latch** it on · switching keeps it latched
[latch] + [1–8]	select silently, without playing
[1–8] + [1–8]	**copy sound** to an empty sound
[latch] + [clear]	**kill all grains** (blur tails on)
[gate] [next]	play the sound · step one right (normalled)
[select]	0–5 v picks 1–8 · empties skipped

02 scenes · morph — mgr_02

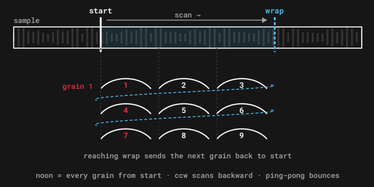
[scene a] [b]	which scene the knobs edit (blinks)
fader	a ← → b · you always hear the fader
[scene] + [scene] / [1–8]	**copy scene** to the other one or a sound
[rand] / [clear] + [scene]	randomize / reset that scene
[morph] cv	±5 v = full sweep · knob attenuverts
shape · reverse	don't blend: each grain picks a or b by chance

03 position — mgr_03

[start]	where grains begin in the sample
[wrap]	how far scan travels before wrapping
[scan]	noon = still · cw forward · ccw back
resets	[1–8] press or gate = back to start
live sounds	start cw = the newest audio coming in
[reverse] long	**ping-pong**: bounce, don't wrap (fades)

__fig.01 : the scan

start · wrap · scan



reaching wrap sends the next grain back to start

noon = every grain from start · ccw scans backward · ping-pong bounces

04 grain — mgr_04

[rate]	0 hz (one grain per trigger) – 130.8 hz
[size]	20 ms – 4 s · noon ≈ 1 s
[pitch]	±2 oct (±3 with mod) · green = octaves
[tone]	ccw lowpass · noon open · cw highpass
[level] [blur]	grain volume · reverb send + decay
[reverse]	grains play backwards (per scene)



tukey = default · discrete: morph picks per grain by chance

05 links · sync — mgr_05

[rate–size]	size sets the rate · rate = 1× – 8×
[size–pitch]	size = audio amount: octave up = half
alt › [sync]	rate ×/÷ 2 4 8 16 of clock · noon ×1
sync + [rate–size]	size snaps to clock divisions



rate–size linked: size sets the base rate, rate multiplies it

07 clear · control all — mgr_07

[clear] › item ×2	**safe clear** · red = clearable
[clear] + item	quick clear, no confirm
items	sound (×1 settings, ×2 sample) · scene · mod · sample memory · load = init · quant
[scene] + turn	value goes to both scenes
[latch] + turn	every sound · + scene = everything
[rand] + [1–8]	randomize a whole sound

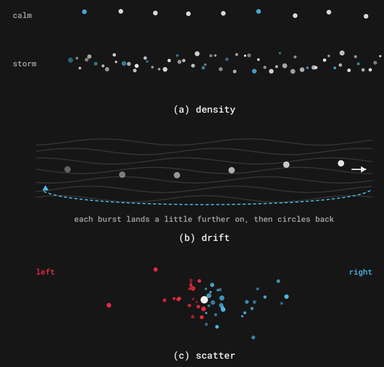
06 modulation — mgr_06

[rand] [x] [y] [z]	mod page · hold = momentary
mod page	knobs attenuvert: cw + yellow, ccw – purple
[rand] / [x] + turn	quick assign from home
[x] [y] [z] jacks	±5 v = full range · pitch 100% = 1v/oct
[rand] + [x] / [alt]	randomize x's (or rand's) assigns
[clear] › turn	on a mod page: clear that knob
rand · start	cw spray · ccw = anywhere in wrap
rand · rate	ccw = trigger chance 100% → 3%
rand · level, tone	ccw = random stereo (l / r opposite)

13 sample loader — mgr_13

alt › [/project] [/wavs] [/recs]	load from that folder
[‹prev] [next›]	step · hold = 5 at a time
[1–8]	preview through the sound's grains

__fig.02 : the storm — mm_04



08 sampling — mgr_08

[sample]	rec (red) · again = stop · 32 s max
then	blue sample = save · empty 1–8 = assign
[sample] 1 s	grab the last 32 s · or open trim
trim page	start / end · scene a / b preview · dir · scan = nudge 25–250 ms
[alt] + [sample]	settings: gain 0.5–8.5× · thrsh · norm
[thrsh]	rec waits for the input (sample fades red)
[resamp]	inputs · resample outs · looping (magenta)

09 live · looping — mgr_09

[1–8] + [sample]	**live sound** from the inputs (magenta)
[latch] + [1–8] + [sample]	same, no trigger
[sample]	magenta = recording · cyan = frozen
[sample] 1 s	save the loop · 8–32 s memory
[thrsh]	auto-freezes when the input goes quiet

10 alt page — mgr_10

[alt] / [alt]	grey labels · hold = momentary
alt › [pitch]	fine-tune ±7 semitones
alt › [a] [b]	cycle output modes (see colours)
[1–8] on alt	sound options: sample live · load choke · save gate · page 2
[1–8] + [save] 1 s	save default sound (load = reset)

11 presets · projects — mgr_11

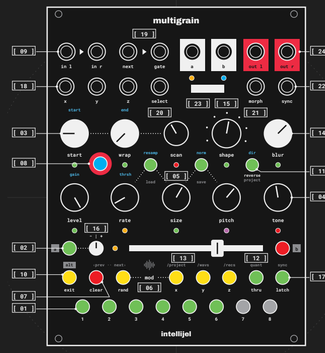
alt › [load] [save] [project]	preset / project pages
[x] [y] [z] › [1–8]	bank (twice = 4–6) › slot
[save] / [load]	quick save · quick reload
[1–8] 4 s	lock preset · collect project
[clear] + [load]	fresh empty preset
room	48 projects × 48 presets × 8 sounds

__fig.03 : front panel

01–13 = sections · 14–24 = hardware

[01] __1–8 : the eight sounds
[02] __scene a / b : scenes, fader
[03] __start wrap scan : position
[04] __grain knobs : six grey knobs
[05] __links : rate–size, size–pitch
[06] __mod : rand · x · y · z
[07] __clear : clear · reset
[08] __sample : record · freeze
[09] __in l / r : audio in · live
[10] __alt / exit : alt page
[11] __reverse : reverse · project
[12] __quant : quantizer
[13] __alt › x y z : sample loader

[14] __blur : reverb
[15] __shape : 7 grain windows
[16] __morph att : cv attenuverter
[17] __thru / latch : monitor · latch
[18] __x y z : cv in ±5 v
[19] __next / gate : step · play
[20] __select : 0–5 v sound pick
[21] __morph : fader cv ±5 v
[22] __sync : clock in
[23] __a / b : cv outs + microsd
[24] __out l / r : audio out



_where things hide

25 · mgr_25

system settings

alt › hold project 4 s

- **rand** random: unipolar (default) / bipolar
- **latch** next jack = scan reset
- **x** auto-collect on preset save
- **start** gate delay, up to 10 ms
- **wrap** select span 5 v › 0.667 v (keyboard)
- **size** max grain: 4 s · ≤ 4 s / sample · ≤ 8 s · full
- **rate–size** shrink grains at high rates
- **clear** 30 s idle = back home (on)
- **thru** knobs jump (default) / catch
- **pitch** a / b 1v/oct trim · size–pitch = test notes

button LEDs

- **lit** on / available
- **dim** off, but can be turned on
- **off** not available here
- **blinking** selected (active sound / scene)
- **rapid** that page is open
- **fading** leads to a page or an action

microsd · boot

- **card** fat32 · 32 gb card included
- **folders** Project01–48 · /Wavs · /Recs
- **latch + clear** hold 4 s = reboot
- **at boot** clear = reset globals · sample / load / save / project = LED theme

LED colours

default theme · no screen, so colour is the display

1–8	loaded · playing · missing · live · live playing	a / b outs	morph · grain trig · grain env · in env · out env · random · pitch · rec · in gate
sample	listening · recording · looping · frozen	firmware	version, shown at boot
mod · pitch	+ mod · – mod · quantized · octave · between		

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[rec] press [rec] hold › then 1 s = keep holding [load] grey = alt label

home page unless marked · alt › = press alt first · knobs edit the blinking sound + scene

samples = .wav · 48_khz · 16-bit · stereo · ≤ 32_sec

red = hold

26 sound options

mgr_26 · alt page · v1.4

alt › [1–8]	the four middle buttons show that sound's options
› [sample]	● **live** · grains from the inputs
› [load]	● **choke** · a new trigger silences every grain already playing
› [save]	● **gate** · grains stop when its gate goes low: gate jack, button or latch
alt › [latch] + [1–8]	set the options for **all** sounds at once
both on	choke + gate = no overlap with other sounds · blur still rings out
keep it	saved with the default sound ([1–8] + [save] 1 s)

29 a / b mod outs

mgr_29 · alt › a / b

alt › [a] / [b]	press again to step through the modes · saved per preset
[alt] + [a] / [b]	quick assign from home
modes	● morph (a) · ● grain trig (b) · ● grain env · ● in env · ● out env
	● random s&h per grain · ● pitch of latest grain, 1 v/oct · ● rec start trig / looping gate · ● input over threshold
pitch out flat?	buffered mult, or trim it in system settings

35 easy to forget

mgr_35 · from the manual + forums

global	system settings, gain, thrsh, resamp, norm, theme, default sounds · in Settings.mgs
per preset	a / b out modes, thru, latch, sync, freeze mod · plus every sound
per sound	sync, links, quantizer, fine-tune, ping-pong, sample
per scene	all knobs, reverse and the mod amounts · the fader morphs them all
fine-tune	alt › pitch · ignores the quantizer · tune pitched samples to c
blur	one global reverb · never choked or gated
rand	unipolar since v1.2 · bipolar is a system setting
select	source under 5 v can't reach sound 8 · scale it with wrap in system settings

27 system settings

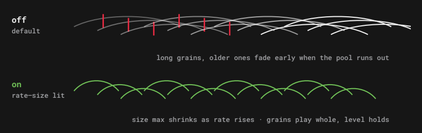
mgr_27 · alt › hold project 4 s

alt › [project] 4 s	open · [exit] = back home · global, not per preset
[rate–size]	● **grain size scaling** · fig.04 · dim = off (default)
[size]	● 4 s fixed · ● ≤ 4 s or sample (default) · ● ≤ 8 s · ● whole sample
[rand]	● unipolar 0–5 v (default) · blinking yellow/purple = bipolar
[latch]	● next jack becomes a scan **reset** (dim = next)
[x] [/project]	● auto-collect samples on preset save
[clear]	● 30 s idle → back home (on by default)
[thru]	● knob catch (pass-thru) · dim = jump (default)
[start]	gate delay, up to +10 ms, for slow cv
[wrap]	select range: 5 v (cw) down to 0.667 v = 8 semitones from 0 v
[pitch] [size–pitch]	a / b pitch-out scale 1.0–1.1× · test octaves 0 / 1 / 2

28 grain size scaling

mgr_28 · fig.04 · v1.4

why	the grain pool is fixed (users report 32) · fill it and something has to give
still	some fading can happen with it on · it just happens less



30 sampling settings

mgr_30 · alt › sample

alt › [sample]	settings only, playback keeps going (sample blinks blue) · or [alt] + [sample]
[gain]	0.5–8.5× · red LED = 1× · line level wants a boost first
[thrsh]	separate for each recorder · ccw = off
› sample rec	rec waits for sound · knob LED blue, red when crossed
› looping	auto-freeze after 1 s of quiet · cyan, magenta when crossed
[resamp]	● dim = ins · lit = resample the outs · ● looping recorder
[size]	looping memory 8 / 16 / 24 / 32 s
[norm]	normalize on save, up to +20 dB · can't be undone

31 trim page

mgr_31 · after recording

[sample] 1 s	open trim (sample fading blue) · the engine pauses
[start] [end]	trim · pass-thru: the knob must cross the value first
[scene a] / [b]	preview from start / the last second · release + hold again to hear new trims
[dir]	dim forward · lit reverse · fading loop
[scan] + [scene a] / [b]	nudge start / end 25 · 75 · 125 · 250 ms (blue fwd, red back)
[sample] / empty [1–8]	save to /recs · or save + assign
red [1–8] ×2	overwrite that sound's sample

32 looping recorder

mgr_32 · live sounds

[1–8] + [sample]	make a live sound · switches the recorder to looping
[sample]	● recording · ● frozen (press to toggle)
[sample] 1 s	save the loop to /recs or an empty sound
no resample	the looping recorder only records the inputs
sync on	live grain starts snap to the clock
last live off	back to the sample recorder, unless set in sampling settings

12 quantizer

mgr_12

alt › [quant] / [quant]	open it (hold quant from home)
keys	clear rand x y z black · 1–7 white
notes	root yellow · scale purple · hold a note key = root
[quant]	knob + mod (dim) / mod only (lit)
[latch] + note	apply to every sound

33 card · files

mgr_33 · where people trip

format	.wav · 48 kHz · 16-bit · stereo · ≤ 32 s
won't work	mono, 24-bit, 96 kHz · 44.1 kHz plays at the wrong pitch · past 32 s is cut off
folders	/Multigrain/Project01–48 · /Wavs (every project) · /Recs · no subfolders
room	128 per project · 128 in /wavs · 1024 recordings
backup	copy the whole /Multigrain folder
card pulled	keeps playing, can't save · reseat, then reboot
no card	ephemeral: 8 samples in ram, gone at power off

34 boot · rescue

mgr_34

[latch] + [clear] 4 s	soft reboot (reindexes the card)
[clear] at boot	reset the global settings
power on + [project]	collect + save every project
[sample] / [load] / [save] / [project] at boot	LED themes: intellijel · reload · golden · royal
version	at boot sample · load · save · project = a.b.c.d

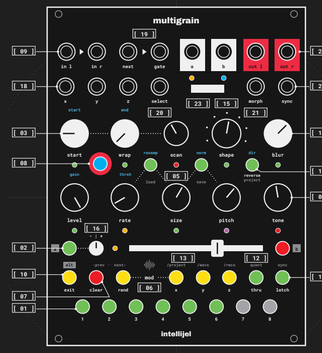
0 1 2 3 4 5 6 7 8 9

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sample	● listening ● recording ● looping ● frozen	firmware	● version, shown at boot
mod · pitch	● + mod ● – mod · quantized ● octave ● between		